

Info

Webpage <https://avik-kar.github.io/>
Github <https://github.com/avik-kar>
LinkedIn <https://www.linkedin.com/in/karavik18/>

Research Interests

My research interests broadly lie in the areas of reinforcement learning, Stochastic approximation, and optimization. Currently, my research focuses on provably efficient reinforcement learning algorithms for Markovian and non-Markovian systems, as well as on convergence guarantees for first- and second-order optimization algorithms.

Education

- Aug, 2021– July, 2026 **Indian Institute of Science, Bangalore**
PhD in Engineering, Dept of Computer Science and Automation , *Current course CGPA – (Tentative) 9.2/10*, Advisor: Prof. Shalabh Bhatnagar
◦ **Fellowship awarded: Prime Minister's Research Fellowship** from Aug 2022 to Jul 2025
- Aug, 2019 – July, 2021 **Indian Institute of Technology Kharagpur**
M.Tech in Electrical Engineering with specialization in Instrumentation and Signal Processing, *CGPA – 9.15/10*
◦ **Thesis:** Federated Learning of Deep Neural Networks with Weakly Labelled Private Datasets
◦ **Award: Keshab K Parhi Endowment Prize** for **best application-oriented thesis** among M.Tech. graduating students of EECS division
- 2013—2017 **St. Thomas' College of Engineering and Technology, Kolkata**
B.Tech in Electronics and Communication Engineering , *Score – 8.54/10*
◦ **Thesis:** Microstrip Line Implementation of X-band Microwave Filter

Publications and Preprints

1. **Policy Gradient Methods for Non-Markovian Reinforcement Learning** [link](#)
Authors: **Avik Kar**, Siddharth Chandak, Rahul Singh, Soumitra Sinhaajari, Eric Moulines, Shalabh Bhatnagar, Nicholas Bambos
In Review.
2. **High-Probability Bounds for SGD under the Polyak-Lojasiewicz Condition with Markovian Noise** [link](#)
Authors: **Avik Kar**, Siddharth Chandak, Rahul Singh, Eric Moulines, Shalabh Bhatnagar, Nicholas Bambos
In Review.
3. **Policy Zooming: Adaptive Discretization-based Infinite-Horizon Average-Reward Reinforcement Learning** [link](#)
Authors: **Avik Kar** and Rahul Singh
Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 40. No. 27. 2026.

4. **Provably Adaptive Average Reward Reinforcement Learning for Metric Spaces** [link](#)
 Authors: **Avik Kar** and Rahul Singh
Proceedings of the Forty-First Conference on Uncertainty in Artificial Intelligence. 2025.
5. **Fantom: Federated Adversarial Network for Training Multi-Sequence Magnetic Resonance Imaging in Semantic Segmentation** [link](#)
 Authors: Anupam Borthakur, Apoorva Srivastava, **Avik Kar**, Dipayan Dewan, and Debdeep Sheet
International Conference on Image Processing (ICIP). IEEE, 2024.
6. **Linear Bandits With Side Observations on Networks** [link](#)
 Authors: **Avik Kar**, Rahul Singh, Fang Liu, Xin Liu, and Ness B. Shroff
IEEE/ACM Transactions on Networking, 2024.
7. **Finite Time Logarithmic Regret Bounds for Self-Tuning Regulation** [link](#)
 Authors: Rahul Singh, Akshay Mete, **Avik Kar**, and P R Kumar
41st International Conference on Machine Learning (ICML), 2024.
8. **Federated Learning for Site Aware Chest Radiograph Screening** [link](#)
 Authors: Arunava Chakravarty, **Avik Kar**, Ramanathan Sethuraman, and Debdeep Sheet
18th International Symposium on Biomedical Imaging (ISBI). IEEE, 2021.

Teaching

- Spring, 2025 **Online Learning and Bandit Algorithms**. Role: Instructor.
- Fall, 2024 **Reinforcement Learning**. Role: Teaching Assistant. Course link: <https://archive.nptel.ac.in/courses/106/106/106106143/>
- Fall, 2023 **Bandit Algorithm (Online Machine Learning)**. Role: Teaching Assistant. Course link: <https://archive.nptel.ac.in/courses/106/101/110101145/>
- Spring, 2023, 2024 **Markov Decision Processes**. Role: Instructor. Course link: <https://avik-kar.github.io/ta/index.html>
- Fall, 2022 **Random Processes**. Role: Teaching Assistant. Course link: <https://ece.iisc.ac.in/~parimal/2022/random/>

Talks and Presentations

- April, 2025 Delivered a talk titled "Provably Adaptive Average Reward Reinforcement Learning for Metric Spaces" at **EECS Research Symposium**, IISc. Slides

Coursework

Optimization, Random Processes, Reinforcement Learning, Statistical Learning Theory, Online Learning, Analysis, Measure Theory, Queueing Theory, Stochastic Control, Statistical Signal Processing, Matrix Theory, Linear Algebra

Technical Skills

- Programming Languages Proficient: Python; Familiar: Matlab, C++, Java
- Software Packages $\text{\LaTeX}$, Pytorch, OpenAI Gym, Mujoco, Scikit-learn

Awards and Achievements

- 2022 Received **Prime Minister's Research Fellowship** from Aug, 2022 to Jul, 2025.
- 2021 Received **Keshab K Parhi Endowment Prize** for **best application oriented thesis** among M.Tech. graduating EECS students of IIT Kharagpur.
- 2020 Ranked 216 (99.74%ile) in GATE, Paper: EC

Professional Experiences

Aug 2017 – **Nomura Research Institute Financial Technology India**
May 2018) Role: Associate Software Engineer